

Seung-Yun Baek

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[Google Scholar Page](#)

Education

Ph.D. in Ecology	2022-2025
Tokyo University of Agriculture and Technology, Japan Thesis: Asian black bears' movement patterns and the influences of landscapes (Advisors: Shinsuke Koike)	
M.S. in Ecology	2020-2022
Tokyo University of Agriculture and Technology, Japan Thesis: What factors affect the distribution expansion of large mammals in central and southern parts of Japan? (Advisors: Shinsuke Koike)	
B.A. in Biology	2013-2019
Jeju National University, Republic of Korea	

Academic Appointments

Assistant professor (Postdoc) , Institute of Global Innovation Research, Tokyo University of Agriculture and Technology, Japan	2025+
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Peer-Reviewed Publications

Published/In Press

- **SY Baek**, A Zedrosser, K Yamazaki, A Shimazaki & S Koike. "Delineating seasonality by inferring behavioral states from movement patterns in Asian black bears" *Mammalian Biology* (2026)
 - N Takekoshi, ..., **SY Baek**, ..., & K Yamazaki. "Relationship between testosterone and movement patterns in male Asian black bear (*Ursus thibetanus*)" *Ecology and Evolution* (2026)
 - K Selvarajah, **SY Baek**, H Takada, D Kim, Y Yamada & S Koike. "Variation in the activity patterns of Japanese serow (*Capricornis crispus*) across different temporal scales" *European Journal of Wildlife Research* (2025)
 - M Shimozuru, ..., **SY Baek**, ..., & H Ito. "Epigenetic clock in bears: A simple cost-effective blood DNA methylation-based age estimation method applicable to multiple bear species" *Ecology and Evolution* (2025)
 - **SY Baek**, A Zedrosser, K Yamazaki, Y Goto, N Takekoshi, & S Koike. "Risky behavior of Asian black bears differs between sex and season in a landscape fragmented by roads" *Journal of Zoology* (2025)
 - **SY Baek**, T Amano, M Akasaka, & S Koike. "The range of large terrestrial mammals has expanded into human-dominated landscapes in Japan" *Communications Earth & Environment* (2025)
 - C Grilo, ..., **SY Baek**, ..., & P Colombo. "Global Roadkill Data: a data set on terrestrial vertebrate mortality caused by collision with vehicles" *Scientific data* (2025)
 - N Maeda, **SY Baek**, K Tochigi, T Naganuma, & S Koike. "Feeding Habits of Japanese Serow in a Warm-Temperate, Low-Elevation, Evergreen Broad-Leaved Shrub-Dominated Forest" *Mammal Study* (2024)
 - **SY Baek**, A Shimazaki, A Zedrosser, T Naganuma, K Yamazaki, & S Koike. "Response to human-modified landscape of an apex consumer: Sex-and season-related variations in road selection and movement pattern" *Global Ecology and Conservation* (2023)
 - M Tanaka, **S Baek**, K Tochigi, T Naganuma, A Inagaki, BS Dewi, & S Koike. "Conditions affecting ant nesting in stumps in a temperate coniferous planted forest" *Forest Ecology and Management* (2023)
 - S Osugi, **S Baek**, T Naganuma, K Tochigi, ML Allen, & S Koike. "The effect of decreasing human activity from COVID-19 on the foraging of fallen fruit by omnivores" *Ecology and Evolution* (2022)
 - S Chowdhury, ..., **SY Baek**, ..., & Tatsuya Amano. "Growth of non-English-language literature on biodiversity conservation"
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Conservation Biology (2022)

- **SY Baek**, T Iwasaki, K Yamazaki, T Naganuma, A Inagaki, K Tochigi, ML. Allen, C Kozakai, & S Koike. “Factors affecting pre-denning activity in Asian black bears” *Mammal Study* (2021)
- **SY Baek**, MH Lee, YS Kim, SR Bae, UR Song, & CK Kang. “Selective oviposition by Oriental fire-Bellied Toads in temporally fluctuating environments” *Current Herpetology* (2021)
- **SY Baek** & SM Lee. “Temporal Patterns and Hotspot Identification of Vehicle Collisions with the Roe Deer (*Capreolus pygargus*) on Jeju Island, South Korea” *Pakistan Journal of Zoology* (2021)
- H Park, A Lim, TY Choi, **SY Baek**, EG Song, & YC Park. “Where to spot: individual identification of leopard cats (*Prionailurus bengalensis euptilurus*) in South Korea” *Journal of Ecology and Environment* (2019)

Under review

- T Makino, **SY Baek**, K Yamazaki, M Allen, & S Koike. “Experience-related variation in search strategies during resource pulse-driven foraging excursions of Asian black bears”

In preparation

- **SY Baek**, T Yamada, K Ando, S Sells, T Tsubota, & S Koike. “Identifying conservation challenges of a small, isolated Asian black bear population through space use and movement patterns”
- Y Yamada, **SY Baek**, H Takada, & S Koike. “Habitat selection of Japanese serow (*Capricornis crispus*) in the Alps”
- K Nakamura, **SY Baek**, & S Koike. “Seasonal migration and selection of large herbivores with high route fidelity”
- Y Arayama, **SY Baek**, ML. Allen, & S Koike. “Characteristics of Asiatic black bear's rubbing behavior”

Presentations

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| • University of Montana (Seminar in Wildlife Biology), Missoula, US (Talk Invitation) | Oct 2025 |
| • Expo 2025 (The Future of Earth and Biodiversity Week: Big Data and Biodiversity), Osaka, Japan (Speakers) | Sep 2025 |
| • The 2025 Annual Meeting of the Mammal Society of Japan, Sapporo, Japan (Poster) | Aug 2025 |
| • 28th International Association for Bear Research and Management (IBA) conference, Edmonton, Canada (Talk) | Sep 2024 |
| • The 2024 Annual Meeting of the Mammal Society of Japan, Hyogo, Japan (Poster) | Sep 2024 |
| • The 2023 Annual Meeting of the Mammal Society of Japan, Okinawa, Japan (Poster) | Sep 2023 |
| • 13th International Mammalogical Congress (IMC), Alaska, U.S (Poster) | Jul 2023 |
| • Gordon Research Conference (GRC): Movement Ecology of Animals, Lucca, Italy (Poster) | May 2023 |
| • Gordon Research Seminar (GRS): Movement Ecology of Animals, Lucca, Italy (Poster) | May 2023 |
| • The 2022 Annual Meeting of the Mammal Society of Japan, Mie, Japan (Poster) | Aug 2022 |
| • The 76th Annual Meeting of the Korean Association of Biological Sciences, Seoul, Republic of Korea (Poster) | Aug 2021 |

Research & Work Experience

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| • Visiting Researcher, Michigan State University, US
(Supervisor/Collaborator: Jerrold Belant) | Nov-Dec 2025 |
| • Visiting Researcher, University of Montana, US
(Supervisor/Collaborator: Sarah Sells) | Oct 2025 |
| • Visiting Researcher, University of Washington, US
(Supervisor/Collaborator: Laura Prugh) | Feb 2025 |
| • Visiting Researcher, University of South-Eastern Norway, Norway
(Supervisor/Collaborator: Andreas Zedrosser) | Mar-Jun 2024 |
| • Investigator, National Institute of Ecology, Republic of Korea | 2020+ |

-Research on Reinforcing Ecological Network Connectivity	
• Field technician, National Institute of Ecology, Republic of Korea -Nationwide Natural Environment Survey (Mammals)	2018-2019
• Sergeant, Army, Republic of Korea Military Service	2015-2017
• Field assistant, National Institute of Ecology, Republic of Korea -Demilitarized Zone (DMZ) Ecosystem Survey -Nationwide Invasive Species Survey (<i>Nutria Myocastor coypus</i>)	2015
• Field assistant, The Biodiversity Foundation, Republic of Korea -Distribution boundary survey for Amur goral (<i>Naemorhedus caudatus</i>)	2015
• Field assistant, UNESCO World Natural Heritage Mt. Hallasan research centre, Republic of Korea -Population and density estimation for Roe deer (<i>Capreolus pygargus</i>)	2014
Mentoring	
Hiroto Masaki, B.A.; M.S., Tokyo University of Agriculture and Technology	2025+
Yoshiya Arayama, B.A.; M.S., Tokyo University of Agriculture and Technology	2024+
Kushaal Selvarajah, Ph.D., Tokyo University of Agriculture and Technology	2024-2026
Makino Tamako, B.A.; M.S., Tokyo University of Agriculture and Technology	2023-2026
Grants & Awards	
• Best Presentation Award - The 2024 Annual Meeting of the Mammal Society of Japan	2024
• FLOuRISH Fellowship for Support for Pioneering Research Initiated by the Next Generation (SPRING) - Tokyo University of Agriculture and Technology	2022-2025
• Academical Talented Student Scholarship - Jeju National University	2018